Unit Mini BPS v1.1

SKU:U090-B













Description

Unit Mini BPS v1.1 is a digital barometric pressure sensor unit that uses the QMP6988 barometric pressure sensor and utilizes an I2C communication interface. This product is mainly used for measuring barometric pressure with a relative accuracy of up to ±0.039 hPa (±0.12 hPa, equivalent to ±1 m height difference) and a wide measurement range. Additionally, a temperature sensor with an absolute accuracy of 2°C is integrated within the chip.

Features

- Built-in temperature and pressure sensor
- High precision (Pressure accuracy: ±0.039hPa, Absolute temperature accuracy: 2°C)
- Low current consumption: 21.4uA
- Development platforms: Arduino, UIFlow (Blockly, Python)
- 2 x LEGO compatible holes

Includes

- 1 x Unit Mini BPS v1.1
- 1 x HY2.0-4P Grove connection cable (5cm)

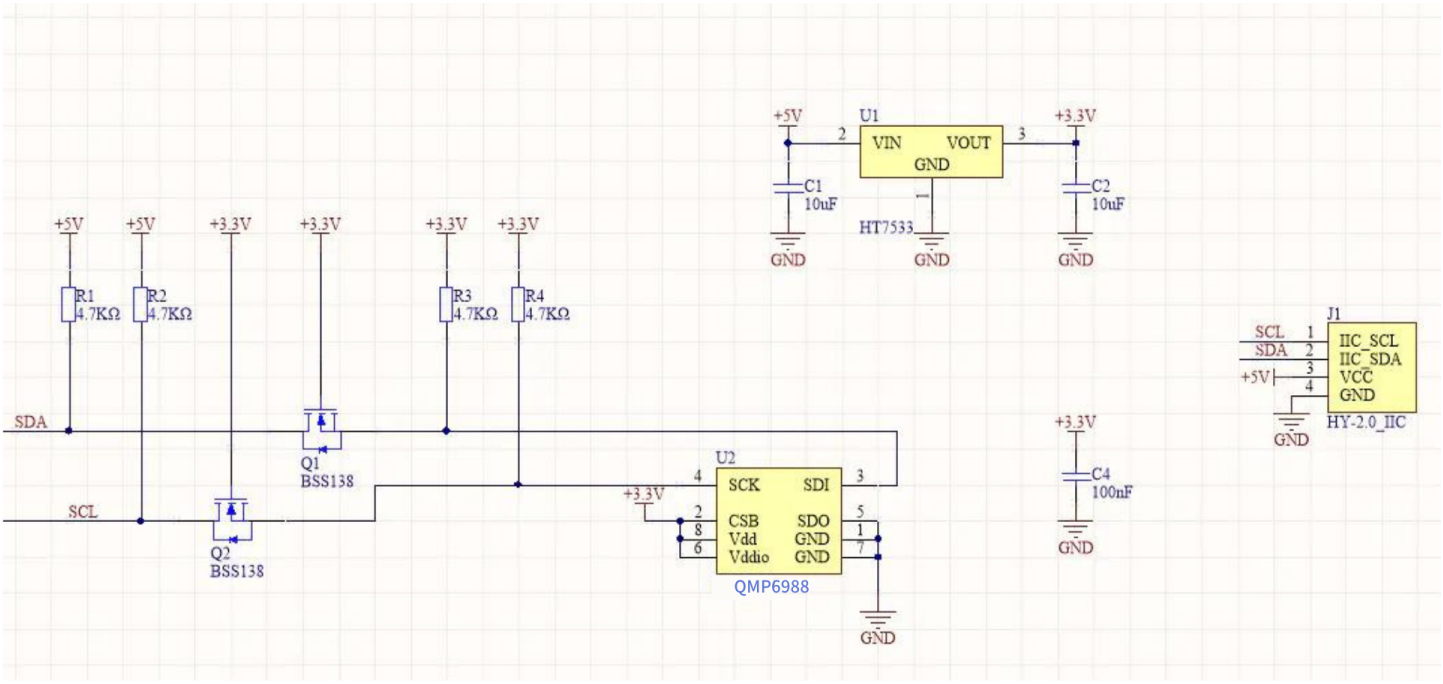
Applications

- Indoor navigation (floor detection)
- Altimeter, height meter
- Weather forecasting

Specifications

Specification	Parameter
Sensor	QMP6988
LDO	HT7533
Pressure Range	30kPa to 110kPa
Relative Pressure Accuracy	±3.9
Pressure Resolution	0.06Pa
Temperature Resolution	0.0002°C
Temperature Range	-40 ~ 80°C
Communication Interface	I2C @0x56
Operating Voltage	5v
Product Size	24.0 x 24.0 x 8.0mm
Product Weight	3.3g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	7.0g

Schematics

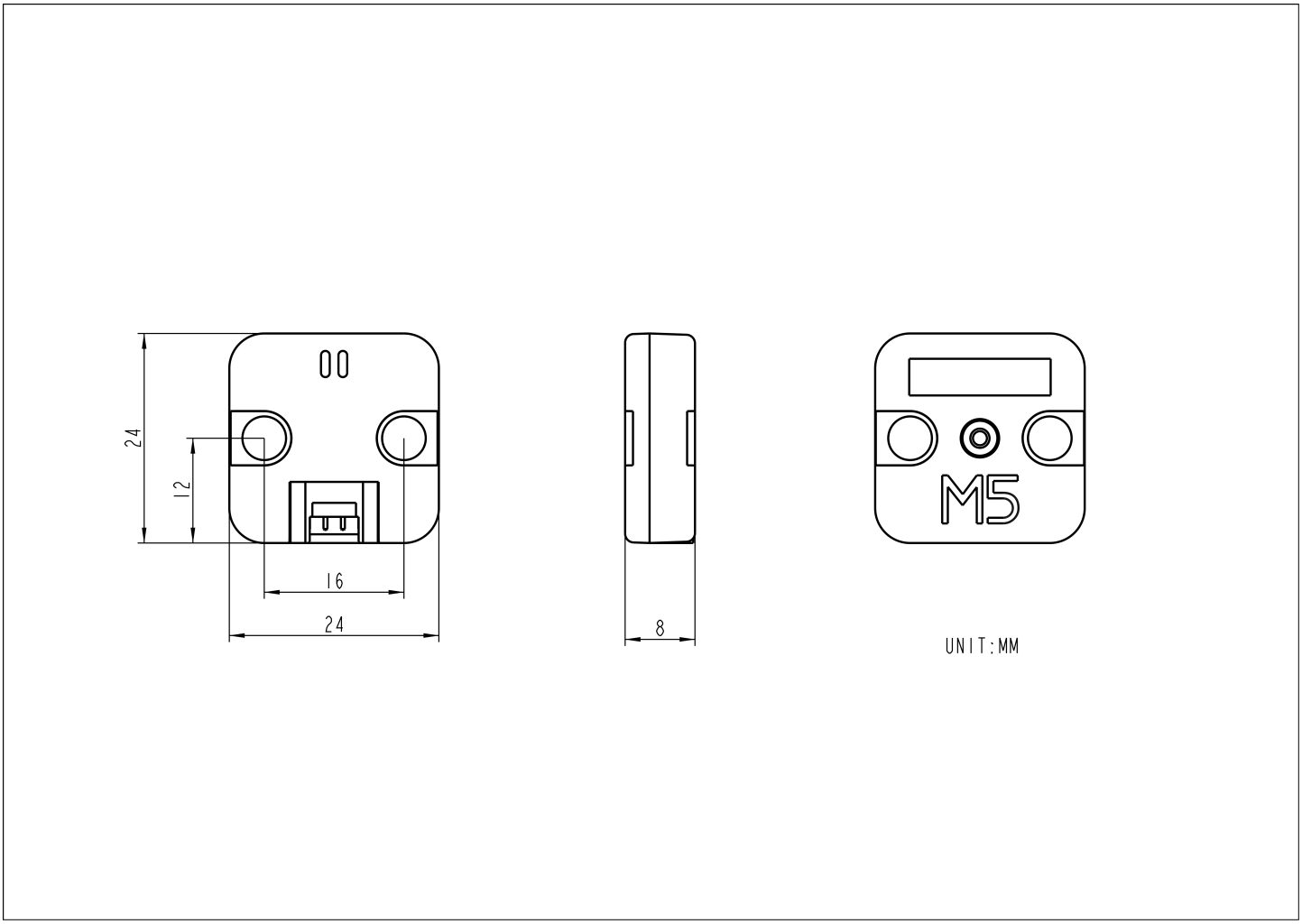


PinMap

Unit Mini BPS v1.1

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Structure

- Unit Mini BPS v1.1 Structure Files

Datasheets

- QMP6988

Softwares

Arduino

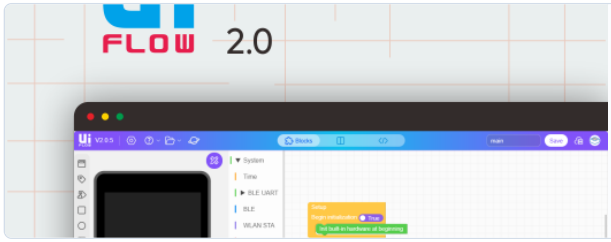
- Unit Mini BPS v1.1 Arduino Tutorial
- Unit Mini BPS v1.1 Example with M5Core
- Unit Mini BPS v1.1 Example with M5Atom
- Unit Mini BPS v1.1 Example with M5AtomS3
- Unit Mini BPS v1.1 Example with M5AtomS3Lite
- Unit Mini BPS v1.1 Example with M5Core2
- Unit Mini BPS v1.1 Example with M5StickC
- Unit Mini BPS v1.1 Example with M5StickCPlus



UiFlow1

- Unit Mini BPS v1.1 UiFlow1 Docs

UiFlow2



UiFlow2 Unit bps Docs

How to use Unit bps and related API instructions in the UiFlow2

Video